CFRM 502, Winter 2020.

Homework 5

Due: Monday 2nd March 2020, NO LATER than 11:59pm

Instructions: You must submit your work through Canvas, and in the form of two attachments: A .pdf file of your detailed answers and a file containing your codes (.R or .Rmd). The .pdf files can be typeset or be a readable scan of your handwritten notes. You do not need to include R outputs in the .pdf file. However, you must clearly indicate the relevant part of your R codes when answering a question that is based on your code.

The maximum number of points you can receive for this homework is 55.

1. (a) (3 pts) Derive expressions for 1-steps to 3-steps ahead forecasts for a stationary ARMA(1,1) model.

   (b) (4 pts) Find the expressions of the large-sample approximations for the variance of the forecast errors in part (a).

   (c) (4 pts) Recall the monthly crude oil time series from Homework 4, Question 4. Load the data using the code and data file provided in Homework 4. Let $\{P_t\}$ be the monthly oil prices, and $r_t = \Delta P_t$ be the monthly price changes. In the previous homework, we fit an MA(2) model and an AR(1) model to the time series $\{r_t\}$. Use each model to find up to 12-step ahead (i.e. 1 year into the future) prediction interval for $\{r_t\}$ with forecast origin December 2007. Show the prediction interval in the time series plot (Hint: See output [8] of Lecture 10).

   (d) (4 pts) One way to evaluate the quality of a time series model is to measure its "out-of-sample" predictive power. Consider a time series process $\{r_t\}_{t=1}^{T}$. Recall that $\widehat{r}_t(h)$ denotes the h-step ahead forecast at time t, which estimates r_{t+h} given the information $r_1, \ldots, r_t$. In particular, note that to find $\widehat{r}_t(h)$ we should fit a model to the sample $r_1, \ldots, r_t$, while to find $\widehat{r}_{t+1}(h)$ we should fit a model to the sample $r_1, \ldots, r_{t+1}$. We define the out-of-sample root mean square errors (RMSE) for h-step ahead forecasts as follows

$$RMSE(h) := \sqrt{\frac{1}{T-h-t_0+1}\sum_{t=t_0}^{T-h}\left[r_{t+h} - \widehat{r}_t(h)\right]^2}.$$

   A smaller value of RMSE(h) indicates a better predictive power for the model. For the two models in part (c), find RMSE(1), RMSE(2), and RMSE(3). Assume that t_0 is December 2007 and T is January 2018. Which model has a better predictive power?

2. Consider the monthly new one family houses sold in the U.S. from January 1963 to November 2018. The data can be obtained by the following R code:

```
library(quantmod)
HS = getSymbols("HSN1FNSA",src="FRED", auto.assign=FALSE)
HS = ts(as.numeric(HS), start = c(1963,1), frequency = 12)
```

   Let $HS = \{HS_t\}$ be the time series of house-sold data.

   (a) (1 pts) Find the sample ACF plot of dHS:$= \Delta HS$. What is the periodicity of dHS based on this plot?

(b) (8 pts) Build an Airline model for HS. You should perform model checking. Write down the fitted model. Plot HS with 12-step ahead (i.e. 1 year into the future) forecasts and prediction intervals (Hint: see output [27] of Lecture 10).

(c) (8 pts) Repeat part (b), but this time use "auto.arima()" function with BIC to build a model for dHS:= ΔHS.

(d) (3 pts) Compare the Models in part (b) and (c). Use in-sample comparison based on BIC. Also, use out-of-sample comparison using RMSE(1), RMSE(2),... RMSE(12) that was defined in Question 1 part (d). Assume that t_0 is January 2000 and T is November 2018.

3. Consider the daily simple returns of IBM stock from 1970 to 2008. Use the following code to obtain the data from the text file "HW5-Q3.txt" that accompanies this homework.

```
dat = read.table("HW5-Q3.txt",header=T)
rtn = xts(dat[,2], order.by=as.Date(as.character(dat[, 1]),"%Y%m%d"))
```

(a) (4 pts) Let $a = \{a_t := |\text{rtn}_t|\}$ be **the absolute values** of daily simple returns of IBM stock. Test if $\{a_t\}$ is stationary using ADF, PP, and KPSS tests. Also, compute the first 100 lags of the sample ACF of $\{a_t\}$. Is there evidence of long memory? Why?

(b) (6 pts) Build an ARFIMA (a.k.a. FARIMA) model for $\{a_t\}$. You should perform model checking. Write down the fitted model.

4. Use the following code to obtain the 1 minute (intraday) spot and futures log-prices of S&P 500 index in May 1993 from the text file "HW5-Q4.txt" that accompanies this homework.

```
dat = read.table("HW5-Q4.txt",header=T)
s = dat[,1];f = dat[,2]
```

The futures prices are for the June contract. Let $\{s_t\}$ and $\{f_t\}$ be the spot and futures prices, respectively.

(a) (2 pts) Define $y = \{y_t := \Delta f_t\}$ and $x = \{x_t := \Delta s_t\}$. Show the time series plot and the scatter plot of (x_t, y_t). Are x and y correlated?

(b) (2 pts) Fit a line regression $y_t = \beta_0 + \beta_1 x_t + \varepsilon_t$ using OLS. Write down the fitted model. Is the model spurious regression?

(c) (6 pts) Build an ARIMAX model for y with x as the predictor. Use "auto.arima()" with BIC to find the model. Perform model checking.